

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I Shri Shanmuga Priya K (192324035) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- | | | | |
|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated | | | |

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

Theory

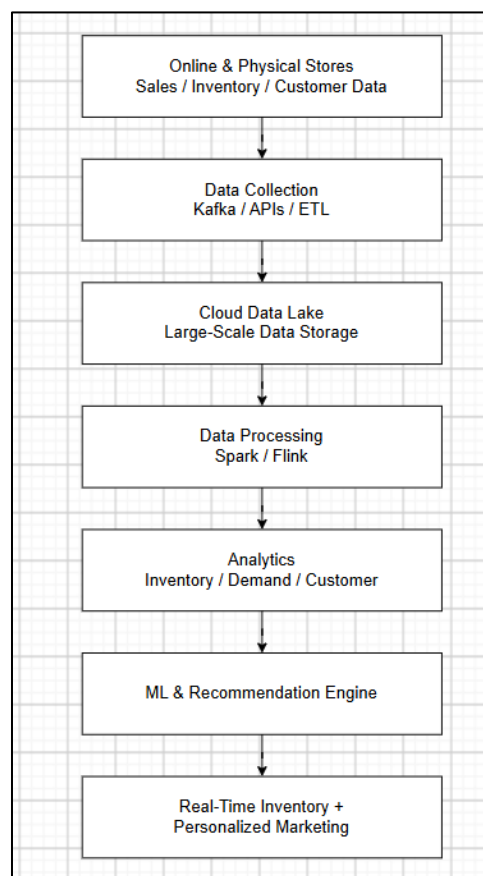
A retail organization generates large volumes of data from physical stores, online shopping platforms, mobile applications, inventory systems, and customer interactions. This data can be used to monitor inventory in real time, forecast product demand, and provide personalized recommendations.

The proposed architecture uses Kafka for real-time data ingestion, a cloud data lake for storing large volumes of structured and unstructured data, and Apache Spark/Flink for distributed and real-time processing.

For demand forecasting, machine learning algorithms such as ARIMA, XGBoost, or LSTM can be used. For personalized recommendations, collaborative filtering, content-based filtering, or hybrid recommendation models can be applied.

Real-time inventory analytics helps identify low-stock products and avoid stock-outs. Demand forecasting helps the organization maintain the appropriate inventory level. Customer behavior analytics helps provide personalized products and offers.

Flowchart



Conclusion

Thus, a big data architecture using Kafka, cloud storage, Spark/Flink and machine learning provides real-time inventory management, accurate demand forecasting and personalized customer recommendations.

2. Social Media Fraud Detection Problem

Theory

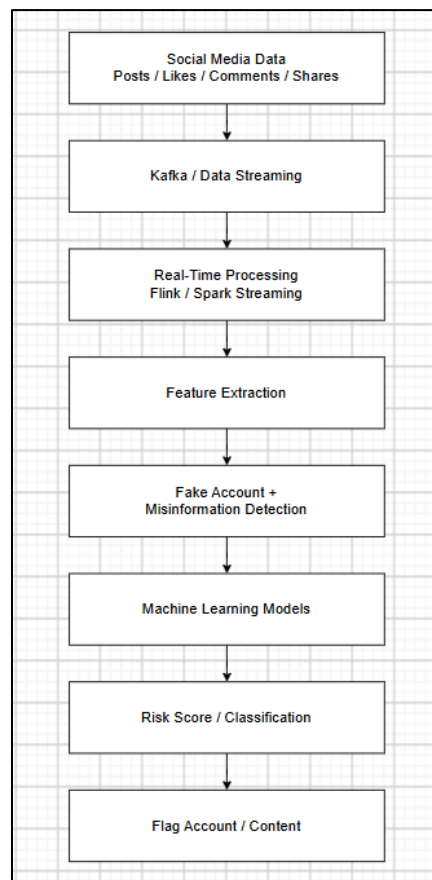
A social media platform generates billions of events such as posts, comments, likes, shares, follows and user logins. Processing this data using traditional batch systems may cause delays in detecting fake accounts and misinformation.

A scalable solution uses Apache Kafka to collect continuous user events. Apache Flink or Spark Streaming processes these events in real time or near real time.

Machine learning models analyze user and content features. Fake accounts can be identified using features such as account age, posting frequency, follower-to-following ratio, device information and unusual activity patterns.

Misinformation can be detected using Natural Language Processing (NLP), text classification and transformer-based models. The system can assign a risk score and automatically flag suspicious users or content for further review.

Flowchart



Conclusion

Therefore, stream processing provides low-latency detection, while machine learning provides intelligent identification of fake accounts and misinformation. The distributed architecture can also scale to billions of social media events.

3. Government Data Platform Problem

Theory

A national government data platform integrates census, economic, demographic and geospatial datasets from different departments. Since different departments may use different formats and standards, interoperability and data integration are major challenges.

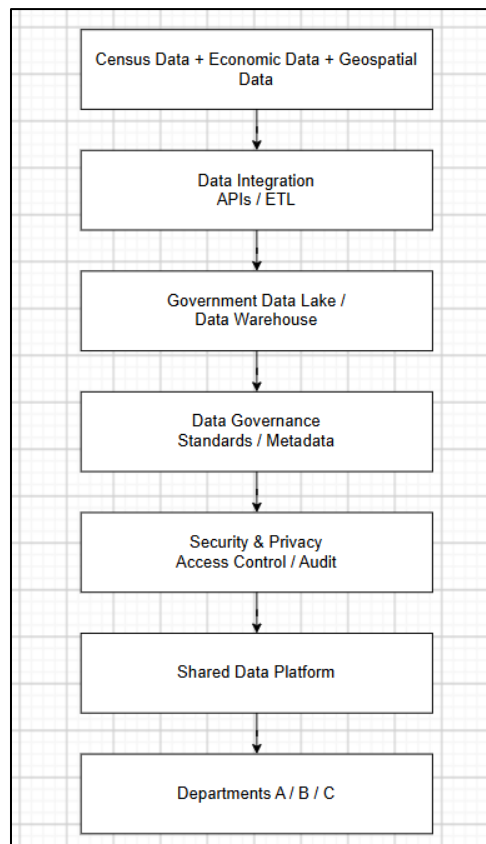
A centralized government data lake/data warehouse can store integrated datasets. APIs and ETL pipelines can be used to collect data from different departments.

A strong data governance framework is required. It should define data ownership, data standards, metadata, access permissions, data quality and security policies.

Common data formats and standardized metadata improve interoperability. A data catalog allows departments to discover available datasets. Master Data Management can ensure that common entities have consistent identifiers.

Because government data may contain sensitive personal information, security mechanisms such as encryption, authentication, role-based access control, auditing and data masking should be implemented.

Flowchart



Conclusion

A government data platform should combine data integration, standardization, governance, security and controlled data sharing. This ensures interoperability while protecting sensitive citizen information.

4. Banking Fraud Detection Problem

Theory

Banks process millions of transactions every minute through credit cards, debit cards, online banking and digital payment systems. Fraud detection must therefore operate with very low latency and high accuracy.

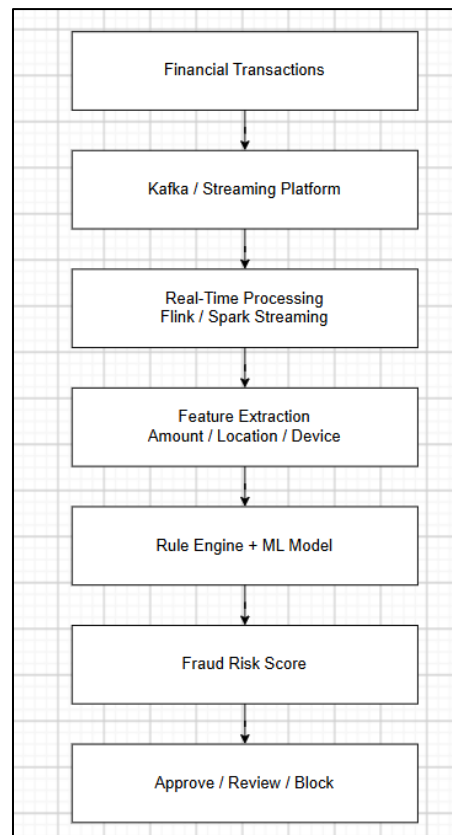
The proposed architecture uses Apache Kafka to capture transaction streams. Apache Flink or Spark Streaming performs real-time processing. Historical transactions are stored in a cloud data lake and are used to train machine learning models.

The system combines a rule engine and machine learning models. Rules can detect known fraud patterns, while machine learning can identify complex and previously unknown patterns.

Algorithms such as Logistic Regression, Random Forest, XGBoost and Isolation Forest can be used. Features such as transaction amount, location, device, time, merchant and transaction frequency are analyzed.

The system produces a fraud risk score. Based on the score, a transaction can be approved, blocked or sent for manual investigation.

Flowchart



Conclusion

The combination of stream processing, historical analytics, rule-based detection and machine learning provides fast and accurate fraud detection while handling millions of transactions at scale.

5. Agriculture Smart Farming Problem

Theory

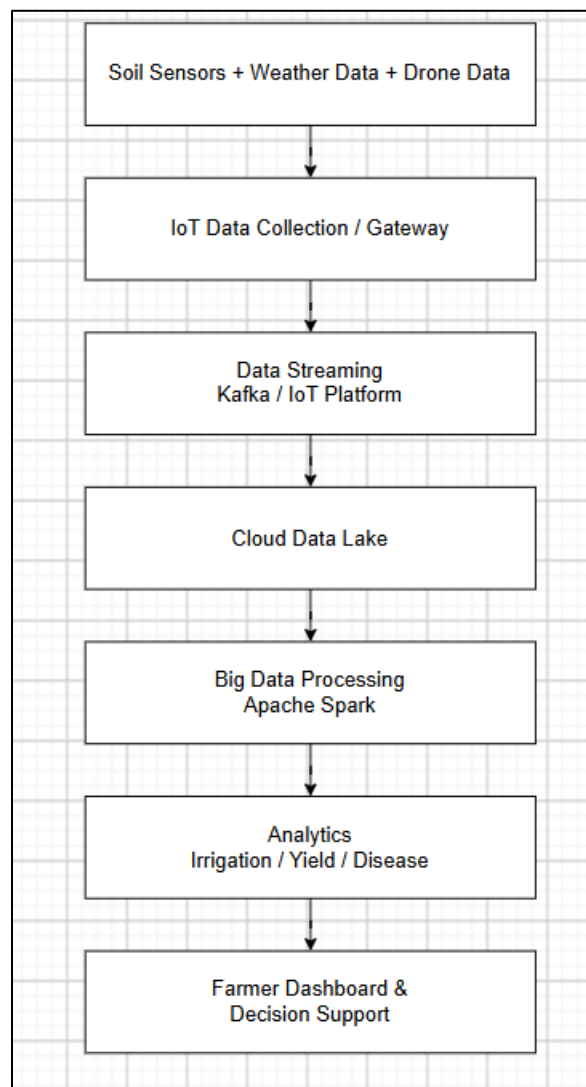
Smart farming uses IoT sensors, drones, weather stations and agricultural databases to collect large amounts of distributed data. Important information includes soil moisture, soil temperature, pH, rainfall, humidity, crop health and weather conditions.

An IoT gateway collects sensor data and sends it to a streaming platform such as Kafka. The data can be stored in a cloud data lake and processed using Apache Spark.

Machine learning and analytics can then be used for crop yield prediction, irrigation optimization, disease detection and resource management.

For example, soil moisture data can determine when irrigation is required. Drone images can be analyzed using computer vision to detect diseases. Historical soil, weather and crop data can be used to predict future crop yield.

Flowchart



Major Analytics Applications

1. Irrigation Optimization

Soil Moisture Data



Moisture Level Analysis



Irrigation Required?



Water Supply Decision

This reduces unnecessary water consumption.

2. Crop Yield Prediction

Historical crop data, weather conditions and soil properties can be analyzed using Random Forest, XGBoost or regression models to estimate crop yield.

3. Disease Detection

Drone images can be analyzed using computer vision and deep learning models to identify unhealthy crops at an early stage.

4. Resource Optimization

Analytics can help optimize:

- Water
- Fertilizer
- Pesticides
- Energy

Conclusion

A big data solution based on IoT, Kafka, cloud storage, Spark and machine learning enables real-time agricultural monitoring and helps farmers improve crop yield while reducing resource consumption.